4.3 Related Rates

Name _____ Period ____ Date _____

AP Calculus 4.3 · Student sheet

I can…

- write an equation that relates the changing quantities.
- differentiate an equation with respect to time t.
- solve related rate problems with circles, spheres, ladders, and cones.

Vocabulary · fill in the missing word

Related rates — rates of change of _____ that are related by an equation

Rate with respect to time — $\frac{dx}{dt}$, how fast x _____ per unit of time

Part 1 · Sliding ruler

A 10-cm ruler: x is the distance of the bottom from the book, y is the height of the top.

x (cm)	6	7	8	9
y (cm)				

Part 2 · Notes and examples (fill in each note, then work the example)

Step 1: draw a picture, name the variables, and write the given rates and the _____ rate.

Step 2: write an equation that relates the _____.

Example 1

The radius of a circle increases at 2 cm/s. How fast is the area increasing when r = 10 cm?

Step 3: differentiate both sides with respect to _____.

Example 2

Air is pumped into a spherical balloon at 100 $\frac{cm^3}{s}$. How fast is the radius increasing when r = 5 cm?

Step 4: substitute the values at the instant only _____ differentiating.

Example 3

A 10-ft ladder leans against a wall. The bottom slides away at 1 ft/s. How fast is the top changing when the bottom is 6 ft from the wall?

Example 4

Water pours into an inverted cone (radius 2 m, height 4 m) at 3 m^3/min. How fast is the water rising when it is 2 m deep?

Example 5

$x^2 + y^2 = 25$ with $\frac{dx}{dt} = 2$. Find $\frac{dy}{dt}$ at (3, 4).

Part 3 · Partner practice

Partner 1

The side of a square increases at 3 cm/s. How fast is the area increasing when the side is 5 cm?

Partner 2

(AP® Practice) A spherical balloon is inflated at $32\pi \frac{cm^3}{s}$. How fast is the radius increasing when r = 2 cm?

(A) 2 cm/s (B) $\frac{1}{2}$ cm/s (C) 8 cm/s (D) 4 cm/s